

Gamified Alzheimer's Disease Diagnosis via Virtual Reality

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Figure 1: Gamified AD diagnosis VR environment. Participants complete the following tasks in a shop scenario: A) a shopping list and a pre-recorded voice instruct participants to memorize and fetch required items in the supermarket (verbal memory); B) participants determine the order of purchase by clicking each item's buttons on the map in their best-perceived order and start shopping (working memory); C) participants encounter distractions like discount banners (executive functions).

ABSTRACT

Virtual reality (VR) has been demonstrated as a promising tool for cognitive assessment thanks to its immersive experience. In this regard, we propose a VR Alzheimer's disease (AD) diagnosis method to allow patients to engage in real-life shopping tasks. We design sub-tasks tailored to the targeting user group to evaluate their cognitive status based on the collected metrics. Our method tries to help assess the status of AD, aiming at a feasible and effective reference for physicians regarding recovery status diagnosis.

Index Terms: Alzheimer's Disease Diagnosis—Virtual Reality—Gamification—Task Design;

1 INTRODUCTION

Alzheimer's disease (AD) is a progressive neural degenerative disorder that irreversibly affects the brain, leading to cognitive decline, memory loss, and consequently poor performance in daily activities. According to Alzheimer's Disease International, more than 55 million people worldwide live with AD¹. Due to the lack of effective cures, the vitality of AD diagnosis and intervention is eminent.

Current AD diagnosis consists of various methods, and no individual method can establish whether an individual has AD or another

form of dementia. Physicians use diagnostic instruments in conjunction with medical history and other information, including cognitive and functional assessments, brain imaging (MRI, CT, PET), and blood tests to make an accurate diagnosis². Among them, cognitive assessment is relatively economical to be repeatedly conducted to keep track of cognitive status. However, current cognitive trials have limited ecological validity [3], because they are always conducted in a carefully controlled environment with fixed instructions without resemblance to cognitive challenges in daily life. Specifically, performance in neuropsychological tests only accounts for 4.6–21.4% variance in patients' daily functioning, according to previous research [4]. Therefore, neuropsychological test outcomes provide little insight into a patient's cognitive status, and it's recommended by a study [2] that cognitive capacity is considered a secondary result in cognitive trials. As a result, current cognitive assessment measures do not directly evaluate the functioning of patients in real life, and this limitation calls for a novel approach that captures patient's cognitive functioning in real-life scenarios.

VR provides a platform for patients to engage in gamified immersive real-life cognitive challenges administered in a controllable environment. Hence, VR holds potential for real-life cognitive functioning assessment with ecological validity [3]. There are studies taking VR as a promising tool in cognitive testing already [1], and they designed a kitchen scenario [3] where participants engage in 5 sub-tasks to do cognitive assessment. But we find the kitchen scenario falls short in two perspectives: 1) 15-minute-long 5-subtask appears to be a burden for potential AD patients; 2) not every one has much real-life experience with kitchen scenarios (especially those who barely cooks).

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¹<https://www.alzint.org/about/dementia-facts-figures/>

²<https://www.dementia.org.au/about-dementia/dementia-research/better-ways-of-diagnosing-dementia>

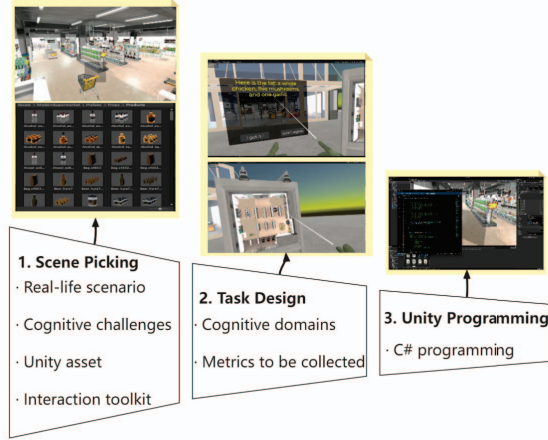


Figure 2: Virtual Reality Development Methodology

Therefore, in this work, we develop an immersive interactive VR supermarket scenario where users are instructed to buy the required items, which is more common to most users. The scenario consists of multiple sub-tasks. It measures several domains of cognition and can store the data online for analysis. This whole process doesn't require any precise medical instrument but a PC with VR headset, and can be conducted periodically within 5 minutes. Our method contributes several advantages as follows.

- After getting used to VR usage, it becomes very easy and economical to keep close track of users' recovery status.
- The user can do the test at any time, without reservation or the commute to the hospital, with nearly zero cost.
- The app is easy to add more exercises to help the user recover, besides diagnosis.

2 METHODS

As depicted in Figure 2, our interactive VR scenario was developed following three stages:

1. Scene Picking: We select a real-life scenario fulfilling: a) it should contain several cognitive challenges, b) it should be a familiar case for the majority of the patient group, and c) it shouldn't be too hard.
2. Task Design: We discuss with experts and select the proper metrics (see Table 1) to be collected.
3. Programming: We implement the scene in Unity and conduct some early tests.

2.1 Environment Setting

We programmed logic and implemented our scene in VR on Unity with multiple VR tools and assets. We developed the software on a PC equipped with AMD Ryzen 7 5700X, 16GB RAM and RTX 3080 GPU, and ran it on HTC Vive Pro Eye to leverage its eye movement tracking function.

2.2 VR Scenario and Metrics Design

During the scene selection and task design, we worked together with experts from a local medical university, and decided to use shopping in a supermarket as our scenario, which resembles the daily life experience for the majority of the targeted patient group,

Table 1: Overview of the Cognition Assessment in VR Gamified AD Diagnosis

cognitive domains	Sub-task measures
Verbal memory	Number of correct items remembered. Times of looking at the prompt (score inverted)
Working memory	Number of matches between planned order and practical order
Execute functions	Time spent staring at distractions

especially the elderly, where patients are expected to execute the given instructions. Through the headset, users are instructed by a pre-recorded voice to carry out three tasks related to buying stuff in the supermarket. As Table 1 summarizes, the app assesses verbal memory, working memory, and executive functions, respectively.

Verbal memory is assessed in task 1 (Fig. 1A), where participants are shown a shopping list and instructed by a pre-recorded voice to remember and fetch them in the supermarket. Performance is measured by the number of correct items remembered and times of checking the prompt (score range: 0-6). Working memory is evaluated in task 2 (Fig. 1B), in which users must plan and determine the order of picking up items on the map by clicking on each item in the order they think is the best. Users are expected to complete shopping tasks in the order they initially planned (score range: 0-3). Executive functions are assessed in task 3 (Fig. 1C), where participants engage in shopping while encountering some distractions in the scene. There are some distractions like discount banners placed in the scene, and participants are not supposed to waste too much time on them when shopping. Therefore, the executive functions are assessed by the time participants stare at banners (score range: 0-5), the less time spent on banners, the better the executive functions.

3 CONCLUSION AND FUTURE WORKS

In this work, we developed a VR scenario where participants execute shopping tasks while their performances are collected to do AD diagnosis for recovery status assessment. Three cognitive domains are being tested with selected evaluation metrics. In the future, we plan to test its validity and feasibility in clinical tests and extend it to cognitive treatment and training by adding extended functions. We also consider taking EEG signals as another evaluation metric to enhance our cognitive status diagnosis.

REFERENCES

- [1] L. D. Hørlyck, K. Obenhausen, A. Jansari, H. Ullum, and K. W. Miskowiak. Virtual reality assessment of daily life executive functions in mood disorders: associations with neuropsychological and functional measures. *Journal of Affective Disorders*, 280:478–487, Feb 2021.
- [2] K. W. Miskowiak, K. Burdick, A. Martinez-Aran, C. Bonnin, C. Bowie, A. Carvalho, P. Gallagher, B. Lafer, C. López-Jaramillo, T. Sumiyoshi, et al. Methodological recommendations for cognition trials in bipolar disorder by the international society for bipolar disorders targeting cognition task force, 2017.
- [3] K. W. Miskowiak, A. E. Jespersen, L. V. Kessing, A. S. Aggestrup, L. B. Glenthøj, M. Nordentoft, C. V. Ott, and A. Lumbye. Cognition assessment in virtual reality: Validity and feasibility of a novel virtual reality test for real-life cognitive functions in mood disorders and psychosis spectrum disorders. *Journal of Psychiatric Research*, 145:182–189, 2022.
- [4] W. Van der Elst, M. P. Van Boxtel, G. J. Van Breukelen, and J. Jolles. A large-scale cross-sectional and longitudinal study into the ecological validity of neuropsychological test measures in neurologically intact people. *Archives of Clinical Neuropsychology*, 23(7-8):787–800, 2008.